

Noe Cochetel, Ph.D.

Computational Biologist | Pangenomics & Graph-based multi-omics | Reproducible, Scalable Workflows

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PROFESSIONAL SUMMARY

From a Ph.D. in Biology, specialized in computational genomics over 9+ years of post-PhD experience, now architecting end-to-end analysis workflows for multi-omics research. Deep expertise in genetics, phylogenomics, and graph-based pangenomics, backed by hands-on training in molecular biology that grounds computational work in real experimental design. Builds pipelines on modern workflow orchestration frameworks, designed from the outset to be reproducible, version-controlled, and able to scale from small pilot studies to large sample sizes. Also maintains a public-facing genomics platform end to end, from data processing and standardization through deployment, and actively uses agentic coding tools to extend engineering skills beyond a traditional biology background. Looking to apply this experimental-computational foundation to building rigorous, scalable bioinformatic tools for a wider research audience.

CORE TECHNICAL SKILLS

Genetics & Phylogenomics: Phylogenetic inference and molecular evolution (RAxML, PAML, OrthoFinder); population/pan-genetics and GWAS (Plink, GEMMA, pan-GWAS via vg tools); comparative and evolutionary genomics across genus-wide, multi-species datasets.

Genome Assembly & Pangenomics: Phased, chromosome-scale genome assembly from long-read (PacBio HiFi) data for diploid and polyploid genomes; Hi-C-based scaffolding (3D-DNA, Juicer); pangenome graph construction (PGGB: wfmask, seqwish, smoothxg; Minigraph-Cactus); graph-based variant calling; pantranscriptomics and panepigenomics.

Programming, Tooling & AI-Assisted Workflows: Nextflow and Snakemake pipeline development for production, reproducible workflows on shared HPC/GPU clusters; containerized execution with Docker and Apptainer, including NVIDIA Clara Parabricks for GPU-accelerated genomics on NVIDIA infrastructure; Python, developed primarily through agentic coding; Unix/Bash and R (proficient); Git/GitHub version control; hands-on use of AI coding agents (Claude Code and similar) for pipeline development, debugging, and analysis; familiar with prompt/context engineering, model evaluation, and agentic tool-use workflows.

Software Engineering Practices: Automated CI for bioinformatics pipelines (GitHub Actions: nf-core lint checks, nf-test suites); nf-core-structured Nextflow (DSL2, JSON parameter schema validation, bundled synthetic test fixtures for end-to-end testing without external data).

Software / Product Delivery: Full-stack development (HTML, CSS, PHP, Ruby on Rails, Apache2) building and maintaining grapegenomics.com, a public genomics data and tools platform, from backend data pipelines through to user-facing deployment.

Genomics & Variant Analysis: Sequence alignment (BLAT, BLAST, BWA, GMAP, minimap2, MUMmer); variant calling/annotation (GATK, Picard, snpEff); ab initio and functional genome annotation (Augustus, BRAKER3, Helixer, InterProScan, HMMER).

Multi-Omics: RNA-seq/transcriptomics (StringTie, Trinity, DESeq2, edgeR, WGCNA); full-length RNA sequencing (PacBio Iso-Seq); single-cell/single-nucleus transcriptomics (alevin-fry/simpleaf, DropletUtils, Seurat); methylation/epigenomics (Bismark, pb-CpG-tools, methylKit); NGS QC and preprocessing (Fastp, Trimmomatic, FastQC/MultiQC).

Data Visualization & Reporting: IGV, JBrowse2; publication-quality and interactive visualization in R (ggplot2, Shiny, flexdashboard) and Python (numpy, matplotlib, plotly); proficient with Claude's data visualization skill, using it hands-on to rapidly produce exploratory and presentation-ready charts.

Wet-Lab Background: Plant molecular biology and physiology (RNA/DNA extraction, RT-qPCR, plant transformation, greenhouse/field experiments) used to design, execute, and validate computational analyses, with direct experience bridging bench work and downstream bioinformatics.

PROFESSIONAL EXPERIENCE

Staff Research Associate III, Computational Genomics

Sep 2023 – Present

- Manage multiple concurrent projects applying multi-omics approaches (pangenomics, transcriptomics, metabolomics, epigenomics) to characterize genetic diversity across the *Vitis* (grape) genus.
- Design and run computational pipelines for genome assembly, methylation, and comparative analysis on HPC/GPU resources, translating experimental and sequencing data into reproducible, reusable Nextflow workflows.
- Lead development and maintenance of a public website (e.g. grapegenomics.com) delivering normalized genomic data and analysis tools to the broader research community, covering the full lifecycle from data processing and standardization through deployment and upkeep.
- Modernized a published haplotype-reconstruction pipeline (HaploSync) into an nf-core-structured Nextflow DSL2 workflow with automated CI, including GitHub Actions for nf-core linting and an nf-test suite backed by bundled synthetic test data.

Postdoctoral Researcher, Plant Genomics

Jun 2019 – Aug 2023

- Led construction and analysis of the first grape super-pangenome: assembled, phased, and scaffolded nine diploid genomes at chromosome scale from long-read data.
- Built a graph-based pangenome from all-vs-all genome alignments (PGGB) to reveal genome evolution patterns and genus-wide trait associations, resolving structural variation invisible to single-reference approaches.
- Developed reusable, openly published bioinformatics pipelines and datasets subsequently adopted by external collaborators across multiple follow-up studies.

Postdoctoral Researcher, Plant Transcriptomics

Sep 2017 – May 2019

- Combined transcriptomics and physiological measurements to characterize drought-tolerance responses across four grapevine genotypes.

Ph.D. Student, Transcriptomics & Molecular Biology

Oct 2013 – Oct 2016

- Applied RNA-seq in a grafted split-root experimental system to characterize rootstock-dependent transcriptional reprogramming in response to nitrogen availability in grafted grapevine.

Master Student, Molecular Biology

Oct 2011 – Jun 2013

- Studied flowering-time regulation in strawberry through functional and expression analysis of florigen/anti-florigen signaling.

EDUCATION

Ph.D. in Plant Biology, University of Bordeaux, France (2016)

M.Sc. in Plant Biology and Biotechnology, University of Bordeaux, France (2013)

SELECTED PUBLICATIONS

Author of 30+ peer-reviewed publications, including in Genome Biology, Nature Communications, PNAS, Molecular Biology and Evolution, and Trends in Genetics. First-author highlights:

- Cochetel N, Cantu D (2026) From structural pangenomes to functional panomics in plants. *Molecular Biology and Evolution*.
- Cochetel N, Vondras AM, Figueroa-Balderas R, Liou J, Peluso P, Cantu D (2025) Phased epigenomics and methylation inheritance in a historical *Vitis vinifera* hybrid. *Genome Biology* 26:392.
- Cochetel N, Minio A, Guarracino A, Garcia JF, Figueroa-Balderas R, Massonnet M, Kasuga T, Londo JP, Garrison E, Gaut BS, Cantu D (2023) A super-pangenome of the North American wild grape species. *Genome Biology* 24:290.
- Cochetel N, Minio A, Massonnet M, Vondras AM, Figueroa-Balderas R, Cantu D (2021) Diploid chromosome-scale assembly of the *Muscadinia rotundifolia* genome supports chromosome fusion and disease resistance gene expansion. *G3* 11:jkab033.
- Cochetel N, Ghan R, Troups HS, Degu A, Tillett RL, Schlauch KA, Cramer GR (2020) Drought tolerance of the grapevine, *Vitis champinii* cv. Ramsey, is associated with higher photosynthesis and transcriptomic responsiveness. *BMC Plant Biology* 20:55.
- Cochetel N, Escudié F, Cookson SJ, et al. (2017) Root transcriptomic responses of grafted grapevines to heterogeneous nitrogen availability depend on rootstock genotype. *Journal of Experimental Botany* 68:4339–4355.

Invited talks and posters at international conferences including Plant and Animal Genome (PAG); full record available on request.